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Intelligent patient monitor mount

Abstract

Data that is derived from a medical device connected to or communicating with, a patient monitor mount, is detected by the patient monitor mount. The data is monitored for events associated with the medical device. The patient monitor mount then determines that the monitored event corresponds to an event. The patient monitor mount generates at least one command for a visualization device to change the data displayed on the visualization device. The command is then transmitted to the visualization device. Related apparatus, systems, methods and articles are also described.

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Background/Summary

RELATED APPLICATION (1) This is a continuation of, and claims priority to and the benefit of, co-pending U.S. application Ser. No. 15/533,644, filed Jun. 6, 2017, which is a national stage application, filed under 35 U.S.C § 371, of International Application No. PCT/US2015/062987, filed Nov. 30, 2015, which claims priority to U.S. Pat. App. Ser. No. 62/093,961 filed on Dec. 18, 2014, the entire contents of which is hereby fully incorporated by reference.

TECHNICAL FIELD

(1) The subject matter described herein relates to medical devices, in particular, to mounts for patient monitors.

BACKGROUND

(2) Patient monitors are used by health care facilities to monitor and display information about a patient, e.g. vital signs, status of connected devices, etc. Patient monitors typically are portable devices that travel with the patient in order to provide continuous monitoring during care. When a patient arrives at a hospital room or other treatment location, the patient monitor is often plugged into or otherwise connected to a patient monitor mount. Patient monitor mounts provide a physical interface for the patient monitor and are generally fixed to the treatment location. Patient monitor

mounts also provide electrical connection to other devices or infrastructure, e.g. power to recharge patient monitor batteries, network connectivity to other medical devices or hospital computer systems, etc.

SUMMARY

(3) In a first aspect, data that is derived from a medical device connected to or communicating with, a patient monitor mount, is detected by the patient monitor mount. The data is monitored for events associated with the medical device. The patient monitor mount then determines that the monitored event corresponds to an event. The patient monitor mount generates at least one command for a visualization device to change the data displayed on the visualization device. The command is then transmitted to the visualization device.

(4) In a further interrelated aspect, there can be electronic communication between a patient monitor and the patient monitor mount, where the patient monitor displays data derived from coupled medical devices.

(5) In yet another interrelated aspect, the command can cause a visualization in the data displayed on the patient monitor to change.

(6) In some variations, the event can be described by a rule set in a visualization configuration. The rule set can describe rules related to events and the patient monitor uses the rule set to generate the command. The command can be an HTML command.

(7) In other variations, the visualization can be triggered based on a device rule set, a phase rule set, and/or an event rule set.

(8) In further variations, the event can be an additional medical device being coupled to, or initiating communication with, the patient monitor mount. In addition or in the alternative, the event can be a change in phase of a medical procedure and/or an occurrence of an alert.

(9) In yet another variation, the visualization configuration can include a list, priority information, positions, and display preferences of diagnostic traces displayed on the patient monitor or visualization device. The data displayed can also be replicated on a display of a computer connected to a network. The connection can be wired or wireless.

(10) In another variation, the patient monitor mount can further include a dock connector that enables a mechanical and electrical connection between the patient monitor and the patient monitor mount. There can be a power supply to provide power to the patient monitor mount. A pod expansion port can be connected to the dock connector. There can be a registered jack connected to the dock connector to enable internet communication with the patient monitor mount. There can be an RFID and WiFi interface to enable RF and wireless communication. There can also be an Ethernet switch, a random-access memory, a persistent memory, and a microprocessor.

(11) Implementations of the current subject matter can include, but are not limited to, methods consistent with the descriptions provided herein as well as articles that comprise a tangibly embodied machine-readable medium operable to cause one or more machines (e.g., computers, etc.) to result in operations implementing one or more of the described features. Similarly, computer systems are also described that may include one or more processors and one or more memories coupled to the one or more processors. A memory, which can include a computer-readable storage medium, may include, encode, store, or the like one or more programs that cause one or more processors to perform one or more of the operations described herein. Computer implemented methods consistent with one or more implementations of the current subject matter can be implemented by one or more data processors residing in a single computing system or multiple computing systems. Such multiple computing systems can be connected and can exchange data and/or commands or other instructions or the like via one or more connections, including but not limited to a connection over a network (e.g. the Internet, a wireless wide area network, a local area network, a wide area network, a wired network, or the like), via a direct connection between one or more of the multiple computing systems, etc.

(12) Implementations of the current subject matter can provide one or more advantages. For

example, the patient monitor mount provides upgraded functionality to the patient monitor. Specifically, allowing preconfigured display settings that cause the patient monitor to display custom diagnostic screen sets. Also, by using a rule set, the patient monitor mount can change what is displayed on the patient monitor in anticipation of, or in reaction to, patient events.

(13) The details of one or more variations of the subject matter described herein are set forth in the accompanying drawings and the description below. Other features and advantages of the subject matter described herein will be apparent from the description and drawings, and from the claims. While certain features of the currently disclosed subject matter are described for illustrative purposes in relation to a patient monitor mount, it should be readily understood that such features are not intended to be limiting. The claims that follow this disclosure are intended to define the scope of the protected subject matter.

Description

DESCRIPTION OF DRAWINGS

- (1) The accompanying drawings, which are incorporated in and constitute a part of this specification, show certain aspects of the subject matter disclosed herein and, together with the description, help explain some of the principles associated with the disclosed implementations. In the drawings,
- (2) FIGS. 1A-C are diagrams illustrating a patient monitor, connected to a patient monitor mount, displaying context-based data;
- (3) FIG. 2 is a block diagram illustrating a visualization configuration used by the patient monitor mount;
- (4) FIG. 3 is a process flow diagram illustrating the data of the patient monitor being updated by the patient monitor mount;
- (5) FIG. 4 is a diagram illustrating a hardware configuration of the patient monitor mount; and
- (6) FIG. 5 is a system diagram illustrating the patient monitor mount connected to an information technology system.
- (7) When practical, similar reference numbers denote similar structures, features, or elements.

DETAILED DESCRIPTION

- (8) The current subject matter is directed to methods, systems, apparatus, articles/computer program products for displaying dynamic or customizable patient diagnostics on a patient monitor. In particular, a patient monitor mount provides a unique clinical display generation capability. The clinical display is automatically generated by monitoring the medical device data on the hospital network. The displayed data can also be configured by caregivers to be customized to the status of medical devices connected to the patient, or to other events. The images from the clinical display can also or alternatively be displayed as a webview by any networked device capable of displaying webpages.
- (9) FIGS. 1A-C are diagrams **100A-C** illustrating a patient monitor **110**, connected to a patient monitor mount **120**, displaying context-based data **130**. The patient monitor mount **120** provides functionality to the patient monitor **110** that allows the patient monitor **110** to display data related to the status of medical devices **140** connected to the patient or in response to patient events. There can be several kinds of contexts, or modes of use, for the patient monitor mount **120** that change the data **130** depending on what is occurring with the patient. For example, there can be an “ensemble” context, a “workflow” context, and an “event” context.
- (10) The ensemble context can relate to the patient monitor mount **120** detecting changes in the status of an ensemble of connected medical devices **140** and updating the data **130** of the patient monitor **110**. For example, as shown in diagram **100A** of FIG. 1A, the patient monitor **110** can be mounted on the patient monitor mount **120**, enabling electronic communication between the patient

monitor **110** and the patient monitor mount **120**. The patient monitor **110** can be any sort of patient monitoring device that can be used to monitor the vital signs/wellbeing or medical device **140** data **130** associated with a patient.

(11) In another implementation, the patient monitor mount **120** need not be connected to a patient monitor. Here, the patient monitor mount **120** can detect data derived from medical devices **140** that are in communication with the patient monitor mount **120**. The patient monitor mount **120** can perform functions, described in greater detail below, to display data on connected visualization devices.

(12) The patient monitor mount **120** can be a physical docking station, described in more detail in FIG. **3**, which connects the patient monitor **110** to the systems of the patient care facility. Once mounted, it can be detected that the patient monitor **110** is mounted to the patient monitor mount **120**.

(13) Once detected, data **130** can be displayed on the patient monitor **110**. The data **130** can include, e.g. diagnostic traces from associated medical devices **140**, patient information, timers, readouts, images, etc. The data **130** can be based on a visualization configuration that is already present in the patient monitor **110**, the patient monitor mount **120**, or a networked computer. The data **130** can be updated in response to a number of scenarios, including e.g. moving the patient, starting a new procedure, updating the visualization configuration, connecting other medical devices to the patient, etc. For example, in FIG. **1A**, after the medical device **140** (in this case an anesthesia device) has been connected to the patient and the network, the patient monitor mount **120**, according to its visualization configuration, updates the data **130** displayed to create a visualization of the updated data. The updated data **130** differs from the prior data **130** in that it includes a diagnostic trace relating to the newly connected medical device **140**. The detection of medical devices **140** and the updating of the patient monitor **110** can be performed by either the patient monitor mount **120** or by a networked computer connected to the patient monitor mount **120**. The patient monitor **110** and the patient monitor mount **120** can be connected to a network **150**, discussed further below, to enable sharing and configuring of data **130** by various users.

(14) The workflow context as illustrated in diagram **100B** of FIG. **1B** can describe the data **130** of the patient monitor **110** being updated according to structured events, i.e. phases. As the patient is treated, the treatment may require the patient to be moved to different locations in the treatment facility. As the patient is moved from location to location, the patient monitor **110** is mounted on the patient monitor mount **120** at each location, which can then update the data **130** of the patient monitor **110** to reflect the visualization configuration at the new location. Another function of the patient monitor mount **120** can be to update the data **130** of the patient monitor **110** according to phases of an operation or procedure or series of procedures. For example, a surgeon may desire a specific set of data to be displayed during the initial phase of a heart transplant operation, and a different set of data to be displayed when the patient is on a cardiopulmonary bypass machine. This functionality can be programmed into the visualization configuration of the patient monitor mount **120** by interfacing an information technology (IT) system **160** with the patient monitor mount **120**. The IT system **160** can include any number of networked computers, servers, databases, etc. needed to manage the visualization configurations used by the patient monitor mounts **120**. One implementation of the IT system **160** is described in FIG. **5**.

(15) The event context as illustrated in diagram **100C** of FIG. **1C** can describe the data **130** of the patient monitor **110** being updated according to unplanned events during patient monitoring. For example, the medical device **140** connected to the patient detects an emergency situation. In addition to sending out an alert, the patient monitor mount **120** can update the data **130** of the patient monitor **110** to display specific data **130** that is needed by caregivers to address the emergency.

(16) FIG. **2** is a block diagram **200** illustrating the visualization configuration **210** used by the patient monitor mount **120**. The visualization configuration **210** used by the patient monitor mount

120 can include a list **220** of diagnostics (or diagnostic traces), e.g. respirator, anesthesia device, etc. The visualization configuration **210** can also include the priority **230** with which to display the diagnostics, i.e. if a choice has to be made between diagnostics to display, which one is more important. There can also be the position **240** of the diagnostics, e.g. first, last, upper left, X-Y coordinates, etc. Also, other preferences **250** can be included, e.g. color, range, scale, refresh rate, etc.

(17) The visualization configuration **210** can also include rule sets to govern events in different contexts, as described above. The visualizations can be triggered by the rule sets in response to events. There can be a device rule set **260** that can display a particular data **130** for one medical device **140**, and another data **130** for another medical device **140**. The device rule set **260** can also govern which medical devices **140** to display data **130** for in the event of a conflict. e.g. too many medical devices **140** connected, etc. There can be a phase rule set **270** that governs displaying data **130** depending on what phase of a procedure is taking place, e.g. where the patient is in the treatment facility, what the current condition of the patient is, etc. There can also be an event rule set **280** that describes what diagnostics are to be displayed on the patient monitor **110** in response to different events. This can include emergency events, where critical information can be automatically displayed as the data **130**. The event rule set **280** can handle situations where the caregiver may not have the time to reprogram the patient monitor **110** to display needed information.

(18) The visualization configuration **210** can also be a combination of user-defined input and predefined input. For example, if during a procedure, the medical device **140** determines that a particular diagnostic trace should be viewed, it can be displayed in an unused frame, presented as a pop-up window, or resizing/reformatting the data **130** so as to display the new information, but without removing information that may have been manually configured by the caregiver.

(19) FIG. 3 is a process flow diagram illustrating the data **130** of the patient monitor **110** being updated by the patient monitor mount **120**. At **310**, the patient monitor can be mounted to the patient monitor mount.

(20) At **320**, the patient monitor mount **120** can detect that the patient monitor **110** has been mounted and that electronic communication has been enabled between the patient monitor **110** and the patient monitor mount **120**. The patient monitor **110** can display data **130** derived from the medical device **140** connected to the patient monitor.

(21) At **330**, the patient monitor mount **120**, after detecting the patient monitor **110**, can monitor for an event associated with the connected medical device **140**.

(22) A number of checks, queries, etc. can be performed either manually or automatically, to determine if the data **130** needs to be updated. The patient monitor mount **120** can determine that the monitored event corresponds to a rule in the rule set in the visualization configuration **210**.

(23) At **340**, a query can be made to determine if the status of one or more medical devices **140** have changed such that, according to the device rule set **260**, the data **130** should be updated.

(24) At **350**, a query can be made to determine if the phase has changed such that, according to the phase rule set **270**, the data **130** should be updated.

(25) At **360**, a query can be made to determine if an event has occurred such that, according to the event rule set **280**, the data **130** should be updated.

(26) At **370**, a query can be made to determine if the visualization configuration **210** has changed such that, according to the new visualization configuration **210**, the data **130** should be updated.

(27) At **380**, if any of the conditions in **340**, **350**, **360**, or **370** were true, then a command can be generated by the patient monitor mount **120** using the rule set. The command, when received by the patient monitor **110**, causes a visualization in which the data **130** displayed on the patient monitor mount **120** changes. The command can be in any computer language or format, e.g. HTML, JAVA, C, C++, etc. If the conditions in **340**, **350**, **360**, and **370** were not true, then the patient monitor **110** can continue to display the current data **130**.

(28) In another implementation, there can be a visualization device, which can be the patient monitor **110**, but can also be any other type of device capable of generating the visualization, e.g. a terminal, wall monitor, mobile device, desktop monitor, dedicated LCD display, etc. At **390**, the command can be transmitted by the patient monitor mount **120** to the patient monitor **110** and/or to the visualization device.

(29) FIG. **4** is a diagram **400** illustrating a hardware visualization configuration **210** of the patient monitor mount **120**. The patient monitor mount **120** can include a dock connector **410**, a power supply **420**, a pod expansion port **422**, a registered jack **424**, a radio-frequency identification (RFID) interface **426**, a WiFi interface **428**, a random-access memory **430**, a persistent memory **432**, a microprocessor **434**, and an Ethernet switch **436**. The patient monitor mount **120** can include any combination of the above elements as well as including additional elements as required to enable the functionality described in the application. The description of examples of alternate computer hardware components that can be used in lieu of, or in addition to, the components described below, can be found after the discussion of FIG. **5**.

(30) The dock connector **410** can be used to physically and/or electrically connect the patient monitor mount **120** to the patient monitor **110**. The dock connector **410** can include snaps, tabs, threaded holes and screws, levers, etc. to create a physical connection between the patient monitor **110** and the patient monitor mount **120**. The dock connector **410** can also include, e.g. exposed leads, tabs, plates, wires, etc. to enable the required electrical connections between the patient monitor **110** and the patient monitor mount **120**.

(31) The power supply **420** can be, e.g. a DC or AC power line/source from the building, or can be a battery power supply. The power supply **420** can also include transformers and/or physical couplers, e.g. 3-prong to 2-prong, 220 VAC to 120 VAC, etc. to allow the patient monitor mount **120** to connect to/from varying types of power outlets.

(32) The pod expansion port **422** can allow various medical devices/physiological sensors to be coupled to the patient monitor.

(33) The registered jack **424** can be any sort of network or electrical jack, e.g. RJ-11, RJ-45, coaxial threaded jack, BNC, LEMO, etc. The registered jack **424** can be used to allow the patient monitor mount **120** to transfer network signals from the network to the microprocessor and/or the patient monitor **110**.

(34) The RFID interface **426** can be used to enable RFID detection and tracking of the patient monitor mount **120**. Each patient monitor mount **120** can have a unique RFID tag that identifies the patient monitor mount **120** to the network or other RFID reader. The RFID tag can be, e.g. active, passive, battery-assisted, etc.

(35) The WiFi interface **428** can be any sort of transceiver that allows wireless communication between the patient monitor mount **120** and a WiFi transmitter, hotspot, wireless router, etc.

(36) The random-access memory **430** can be used to store instructions executed by the microprocessor **434** during operation of the patient monitor mount **120**.

(37) The persistent memory **432** can be memory used to store visualization configurations **210** or other programming instructions for the patient monitor mount **120**, e.g. flash memory, magnetic memory, etc.

(38) The microprocessor **434** can be any sort of microprocessor and/or chipset that allows execution of programmed instructions for the patient monitor mount **120**.

(39) The Ethernet switch **436** can be any switching element that allows the registered jack to communicate with the patient monitor **110** and the microprocessor **434**.

(40) FIG. **5** is a system diagram **500** illustrating the patient monitor mount **120** connected to the network **150**. There can be multiple patient monitor mounts **120** connected throughout a healthcare facility, via the network **150**. The network connectivity can allow the data **130** of the patient monitor **110** to be displayed at any web-enabled display. Also, caregivers can remotely view a medical procedure being performed on a patient connected to the patient monitor mount **120**

according to their own stored visualization configuration **210**, independent of the visualization configuration **210** of the caregiver(s) actually performing the procedure on the patient.

(41) At **510**, the patient can be connected to one or more medical devices **140**, patient monitors **110**, and/or patient monitor mounts **120**. The devices, at the point-of-care, can be connected to a network via the patient monitor mount **120** or any other connection enabling device, e.g. a wired/wireless router.

(42) At **520**, there can be a network device data aggregator that compiles a list or other memory object that contains reference to all connected medical devices **140**, patient monitors **110**, and patient monitor mounts **120**. It is not necessary that the detected medical devices **140** be connected to the patient monitor mount **120**; any networked medical device **140** can be aggregated.

(43) At **522**, the list from the network device data aggregator at **520** can be stored in a medical device **140** data store. The medical device **140** data store can be, e.g. a database, a server, a cluster, etc. that collects information from the network device data aggregator and stores and organizes it for reference by other networked programs.

(44) At **524**, the automatic configurator can be a program responsible for deciding on the contents of the display based on what data is available the automatic configurator supports multiple canned visualization configurations **210** and can also support user-defined visualization configurations **210**. In one implementation, unless a user changes it, the automatic configurator can provide a default display visualization configuration **210** that replicates the displayed data on the medical device **140**. The automatic configurator can provide a visualization configuration **210** description of the webview layout to describe what images are being displayed to the user.

(45) At **526**, the automatic configurator can provide canned or user-generated visualization configurations **210**. Also, the visualization configurations **210** can be shared between users and further customized by the current user of the visualization configuration **210**.

(46) At **528**, the visualization configurations **210** from **526** can be sent to a presentation view creator. The presentation view creator can be a GUI (graphical user interface) or other development environment that allows the visualization configurations **210** to be customized before being displayed on the webview. The presentation view creator can include, e.g. providing lists of available visualization configurations **210**, other medical devices **140** connected to the network **150**, lists of diagnostic traces available through the medical devices **140** and the patient monitor mounts **120**, etc.

(47) At **530**, the webpage viewer can be a webpage, JAVASCRIPT, or other end-user GUI that can display the view created at **528**. The webpage viewer can be used to proof the webviews and display how they will be shown to anyone viewing the webview on their internet browser.

(48) At **532**, the webpage viewer can export webviews to a web server. The web server can be any computer, server, database, etc. that distributes the webviews to the network. The web server can distribute the webviews to connected devices automatically, or in response to user request.

(49) At **540** and **550**, webviews can be downloaded to network connected medical device **140** monitors wirelessly or wired, respectively. The medical device **140** monitors can be, e.g. large screen TV's, computer monitors, terminals, etc. that can then display the requested visualization configuration **210** of medical device **140** diagnostic traces.

(50) One or more aspects or features of the subject matter described herein can be realized in digital electronic circuitry, integrated circuitry, specially designed application specific integrated circuits (ASICs), field programmable gate arrays (FPGAs) computer hardware, firmware, software, and/or combinations thereof. These various aspects or features can include implementation in one or more computer programs that are executable and/or interpretable on a programmable system including at least one programmable processor, which can be special or general purpose, coupled to receive data and instructions from, and to transmit data and instructions to, a storage system, at least one input device, and at least one output device. The programmable system or computing system may include clients and servers. A client and server are generally remote from each other

and typically interact through a communication network. The relationship of client and server arises by virtue of computer programs running on the respective computers and having a client-server relationship to each other.

(51) These computer programs, which can also be referred to programs, software, software applications, applications, components, or code, include machine instructions for a programmable processor, and can be implemented in a high-level procedural language, an object-oriented programming language, a functional programming language, a logical programming language, and/or in assembly/machine language. As used herein, the term “machine-readable medium” refers to any computer program product, apparatus and/or device, such as for example magnetic discs, optical disks, memory, and Programmable Logic Devices (PLDs), used to provide machine instructions and/or data to a programmable processor, including a machine-readable medium that receives machine instructions as a machine-readable signal. The term “machine-readable signal” refers to any signal used to provide machine instructions and/or data to a programmable processor. The machine-readable medium can store such machine instructions non-transitorily, such as for example as would a non-transient solid-state memory or a magnetic hard drive or any equivalent storage medium. The machine-readable medium can alternatively or additionally store such machine instructions in a transient manner, such as for example as would a processor cache or other random access memory associated with one or more physical processor cores.

(52) To provide for interaction with a user, one or more aspects or features of the subject matter described herein can be implemented on a computer having a display device, such as for example a cathode ray tube (CRT) or a liquid crystal display (LCD) or a light emitting diode (LED) monitor for displaying information to the user and a keyboard and a pointing device, such as for example a mouse or a trackball, by which the user may provide input to the computer. Other kinds of devices can be used to provide for interaction with a user as well. For example, feedback provided to the user can be any form of sensory feedback, such as for example visual feedback, auditory feedback, or tactile feedback; and input from the user may be received in any form, including, but not limited to, acoustic, speech, or tactile input. Other possible input devices include, but are not limited to, touch screens or other touch-sensitive devices such as single or multi-point resistive or capacitive trackpads, voice recognition hardware and software, optical scanners, optical pointers, digital image capture devices and associated interpretation software, and the like.

(53) In the descriptions above and in the claims, phrases such as “at least one of” or “one or more of” may occur followed by a conjunctive list of elements or features. The term “and/or” may also occur in a list of two or more elements or features. Unless otherwise implicitly or explicitly contradicted by the context in which it used, such a phrase is intended to mean any of the listed elements or features individually or any of the recited elements or features in combination with any of the other recited elements or features. For example, the phrases “at least one of A and B;” “one or more of A and B;” and “A and/or B” are each intended to mean “A alone, B alone, or A and B together.” A similar interpretation is also intended for lists including three or more items. For example, the phrases “at least one of A, B, and C;” “one or more of A, B, and C;” and “A, B, and/or C” are each intended to mean “A alone, B alone, C alone, A and B together, A and C together, B and C together, or A and B and C together.” Use of the term “based on,” above and in the claims is intended to mean, “based at least in part on,” such that an unrecited feature or element is also permissible.

(54) The subject matter described herein can be embodied in systems, apparatus, methods, and/or articles depending on the desired configuration. The implementations set forth in the foregoing description do not represent all implementations consistent with the subject matter described herein. Instead, they are merely some examples consistent with aspects related to the described subject matter. Although a few variations have been described in detail above, other modifications or additions are possible. In particular, further features and/or variations can be provided in addition to those set forth herein. For example, the implementations described above can be

directed to various combinations and subcombinations of the disclosed features and/or combinations and subcombinations of several further features disclosed above. In addition, the logic flows depicted in the accompanying figures and/or described herein do not necessarily require the particular order shown, or sequential order, to achieve desirable results. Other implementations may be within the scope of the following claims.

Claims

1. An apparatus, comprising: a visualization device; and a patient monitor mount, the patient monitor mount including: a processor; a memory communicating with the processor, the memory storing: a device rule set comprising pre-defined input; and a set of instructions that, when executed by the processor, perform a method comprising: receiving medical device data from a plurality of medical devices; determine a status of an ensemble of the plurality of medical devices connected to a patient associated with the patient monitor mount based on the medical device data received from the plurality of medical devices; transmit one or more commands between the patient monitor mount and the visualization device that, upon receipt of the one or more commands, cause the visualization device to display a visualization configuration triggered by the device rule set and the determined status of the ensemble of the plurality of medical devices; receiving patient medical data from the plurality of medical devices; determining a workflow context or a patient event context based on the patient medical data received from the plurality of medical devices; and transmitting one or more additional commands between the patient monitor mount and the visualization device that, upon receipt of the one or more additional commands, causes the visualization device to display an updated visualization configuration triggered by the device rule set and the determined workflow context or patient event context.
2. The apparatus of claim 1, wherein the visualization device is configured to make a network connection with the patient monitor mount and transmit customized display information over the network connection.
3. The apparatus of claim 1, wherein the device rule set is configured to determine a set of medical data to be displayed from a particular medical device among a plurality of medical devices is displayed.
4. The apparatus of claim 3, wherein the determining the set of medical data includes forming a resolution in the event of a conflict between at least two of the medical devices.
5. The apparatus of claim 4, wherein the set of instructions is configured to determine that the conflict is present.
6. The apparatus of claim 5, wherein the set of instructions is further configured to resolve the conflict to determine which conflicting medical device presents the medical data received therefrom on the visualization device.
7. The apparatus of claim 2, wherein the customized display information is customized to a location.
8. The apparatus of claim 2, wherein the customized display information is a display context.
9. The apparatus of claim 3, wherein the determining a patient event context includes at least one of detection of a physiological event and detection of an occurrence of an alert.
10. The apparatus of claim 2, wherein the device rule set is configured to determine which of the customized display information to transmit between the visualization device and the patient monitor mount.
11. A method comprising: storing, in at least one of a patient monitor mount and a visualization device, customized visualization configurations in association with a device rule set, the device rule set comprising pre-defined input; receiving, by at least one of the patient monitor mount and the visualization device, medical device data from at least one medical device, the patient monitor mount being separate and distinct from the at least one medical device; determine a status of an

ensemble of the plurality of medical devices connected to a patient associated with the patient monitor mount based on the medical device data; generating at least one command to cause the visualization device to display a visualization configuration using the device rule set based on the status of the ensemble to transmit between the visualization device and the patient monitor mount; receiving patient medical data from the plurality of medical devices; determining a workflow context or a patient event context based on the patient medical data; and generating at least one additional command to cause the visualization device to display an updated visualization configuration using the device rule set based on the determined workflow context or patient event context to transmit between the visualization device and the patient monitor mount.

12. The method of claim 11, further comprising transmitting, over a network connection between the patient monitor mount and the visualization device, the generated at least one command and at least one additional command.

13. The method of claim 12, wherein the at least one command comprises customized display information received over the network connection based on the device rule set.

14. The method of claim 11, wherein the device rule set is configured to govern conflicts between two or more medical devices and, responsive to an event of conflict, to determine based upon the device rule set which of the two or more medical devices will display data on the visualization device communicating with the patient monitor mount.

15. The method of claim 14, further comprising determining, by at least one of the patient monitor mount and the visualization device that the conflict between the two or more medical devices has occurred.

16. The method of claim 13, wherein the device rule is set is configured to determine which of the customized display information to transmit between the visualization device and the patient monitor mount.

17. The method of claim 13, wherein the customized display information is customized to a location.

18. The method of claim 13, wherein the customized display information is a display context.
